LaTeX Examples in R Markdown

2026-08-26

Introduction to LaTeX in R Markdown

This document serves as a comprehensive reference guide for embedding LaTeX math notation and formatting inside R Markdown files.

1. Basic Inline and Block Math

Use single `$` for math that flows with your text, and double `$$` to break the math onto its own centered line.

The standard linear model is written as $Y = \beta_0 + \beta_1 X + \epsilon$, where ϵ represents the error term.

We assume the errors follow a normal distribution:

$$\epsilon \sim N(0, \sigma^2)$$

2. Fractions and Sums

Great for statistics, summaries, and mathematical definitions. Notice that wrapping multiple characters in curly braces `{}` keeps them grouped together under subscripts or superscripts.

The sample mean formula is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

The variance can be computed using:

$$\sigma^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n - 1}$$

3. Multi-line Aligned Equations

Use the `aligned` environment to line up equations by their equal signs. The ampersand (`&`) marks the alignment point, and double backslashes (`\\`) create a new line.

$$\begin{aligned} (x + y)^2 &= (x + y)(x + y) \\ &= x^2 + xy + yx + y^2 \\ &= x^2 + 2xy + y^2 \end{aligned}$$

4. Matrices and Vectors

You can create formatted data tables or coefficient vectors using the `bmatrix` structure. Use `&` to separate columns and `\\` to separate rows.

The design matrix $\mathbf{X}$ is structured as:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix}$$

5. Piecewise Functions (Cases)

Use the `cases` environment to write conditional mathematical definitions. Use `\text{}` to safely write plain text inside math mode.

The indicator function is defined as:

$$I_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

6. Greek Letters and Common Operators

Here is a quick-reference list of how common symbols appear when knit:

- **Greek letters:** α, β, γ, θ, λ, μ, σ, Ω
- **Calculus:** $\int_a^b f(x)\,dx$ and $\lim_{x \to \infty} \frac{1}{x} = 0$
- **Relations:** $\leq$, $\geq$, $\neq$, $\approx$, $\propto$, $\in$
- **Logic & Stats:** $\hat{y}$, $\bar{x}$, $\tilde{\mu}$, $\forall$, $\exists$

7. PDF-Only Layout Tricks

If you are knitting directly to a PDF, you can inject raw LaTeX layout commands like page breaks or vertical spacing directly into your markdown text:

This text sits at the top of the section.

This text has exactly 1.5 centimeters of vertical blank space added right above it.

To force the remaining content to a brand new page, you can use the command: `\newpage`.